DATA SYSTEMS

AlignPro

Lidar boresight calibration (patch test)

STAGE 02 OF THE DCS CHAIN · POSPRO → ALIGNPRO → CLOUDPRO

User Guide

For AlignPro 1.10.1 · November 2026

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1 Introduction

AlignPro finds the mounting angles of the Lidar on the Canadian Lidar mobile mapper (roll, pitch and heading) from a short calibration drive, the **patch test**. It hands the result to CloudPro, the next stage of the processing chain, and gives you the three angles in the mapper's own convention so you can enter them in the mapper. This guide is for AlignPro 1.10.1.

What a patch test is

Each Lidar point is turned into a map coordinate with the trajectory (the vehicle's position and attitude), the **lever arm** (where the Lidar sits relative to the Certus reference point, measured with a tape or total station) and the **mounting angles** (how the Lidar is turned relative to the vehicle). The angles cannot be measured to the hundredth of a degree a point cloud needs: an error of 0.1° moves a point 10 m away by about 1.7 cm, in different directions on passes driven in opposite directions.

A patch test drives past the same solid surfaces (walls, building corners, poles, open ground) several times, in opposite directions and at different distances. With the wrong angles the passes do not agree: a wall seen from both sides of the road appears twice, a little apart. AlignPro finds small flat **patches** that two or more passes see, and solves the three angles that bring the points of every pass onto the same planes (a point-to-plane adjustment). This correction to the recorded mounting is the **boresight**. With three or more passes it also solves with each pass left out in turn; the spread of those answers is the **SDEV** of each angle.

AlignPro in the DCS chain

AlignPro is stage 02 of the Data Collection Systems (DCS) chain. It reads the captures (`.clraw`) and PosPro's trajectory, and finalizes `alignment.json` for CloudPro when you **apply** values; calibrating alone finalizes nothing (Chapter Finalizing).

What's new

VERSION	CHANGES
1.10.1	Sonar heads are named by the side their fan looks to. The Lidar mounting convention comes from DCS Core's shared coordinate frames. Needs DCS Core 1.15.0.
1.10.0	**Sonar patch test:** a marine job opens on one page that solves, per sonar head, the roll, pitch, heading and (when asked) latency corrections that make the seabed agree between passes and between the two heads, keeps a correction only when the survey lines show it, and finalizes them for CloudPro (Marine jobs). Needs DCS Core 1.14.0.
1.9.0	**One calibration flow:** the Calibration page reads `1 AUTO CALIBRATE` › `2 CHECK / FINE TUNE…` › `3 APPLY ADVISED & FINALIZE`, and shows the recorded, advised and in-use angles side by side with whether the advised ones are in use and whether the ones in use are finalized; the second, typed-angles Apply is gone (type values in the Data View). The Data View has `AUTO CALIBRATE` and `APPLY & FINALIZE THESE VALUES`. Every button that applies also finalizes and says so. Continue goes to StripPro when it is installed.
1.8.2	A pair measured across recording holes says *Recording holes* on the Pairs page instead of calibration or trajectory.
1.8.1	**Opened from CloudPro:** `OPEN IN ALIGNPRO` in CloudPro's LINE SEPARATION tab brings up the Pairs page with that line pair selected under *From the point cloud*. Needs DCS Core 1.12.0.
1.8.0	**From the point cloud:** what CloudPro's line separation check says about the calibration, per line pair, on the Pairs page, and `USE THESE LINES AND THEIR OVERLAP` to calibrate on that material (From the point cloud). Needs CloudPro 2.2.1 for the findings.
1.7.0	The new DCS header: the chain and AlignPro's phases (Prepare › Calibrate › Report) as breadcrumbs, with the guidance in their tooltips instead of a sentence under the page. The window and the Data View are always windows of their own (no docking); `OPEN THE DATA VIEW` and `RESET WINDOW` are in the project menu. *Publish* is now *Finalize*. Needs DCS Core 1.11.0.
1.6.2	Packets recorded while the vehicle is slower than the minimum pass speed (Settings, *Passes and pairs*) are left out while reading, so parking and waiting no longer use up the point budget of a line; the moving passes get those points instead. Lines are read again once.

1.6.1	**Point density** on the Calibration page (section Point density): Standard, Dense (now the default for new projects), Very dense or Custom, with an estimate of the memory it needs on this PC; a density that would not fit is refused. Fixes: Apply or a system profile during the full-density read waits for it instead of half-applying; opening a project while a job runs waits for it; STOP no longer shows as an error or saves a partial calibration; holes longer than 1 s in the trajectory are no longer bridged (their points are left out); the Data View places only the points it shows (faster nudges, less memory); drawing tools give the previous map tool back. Needs DCS Core 1.8.0.
1.6.0	**System profiles.** `SYSTEM PROFILE…` on the Installation page (and `SAVE TO SYSTEM PROFILE…` on the Calibration page) keeps the applied Lidar lever arm, mounting angles and latency as the profile of the mapper system that recorded the lines, on this computer. When you open a job from a system that has a profile whose values differ, AlignPro offers to use them (section System profiles). PosPro keeps the antenna offsets in the same profile. Needs DCS Core 1.7.0.
1.5.0	Guided steps: a `‹ BACK` / `NEXT ›` bar under the pages says whether each step is done and what to press, and `CONTINUE IN CLOUDPRO ›` on the last step once the calibration is applied and published. The version number is shown after the plugin name in the header. Needs DCS Core 1.6.0.
1.4.4	One SBET per capture: the trajectory can be several SBET files (older MapPro captures, where each Start/Stop made its own project folder and SBET); each capture uses the SBET that covers it, and nothing is interpolated between them.
1.4.3	Data View: the views follow the inspection boxes (drawn on the map or in the Data View), the area of interest, the plan (`LOAD VIEW AREA`) or the QGIS map (`LOAD MAP EXTENT`); the loaded area is outlined on the map. Inside the boxes every recorded point is shown, re-read from the raw data. Four views at once: plan, roll, pitch and heading.
1.4.2	Fixes: no crash in the Data View after Apply, publishing or a new area of interest when the view area holds no points; fewer Python warnings.
1.4.1	Mapper 2.10 projects: the survey lines are recognised from what the files record (new file names, a line split over several files, older captures grouped into `lines/`); navigation recordings are left out, also when PosPro processed them. Needs DCS Core 1.4.0.
1.4.0	The same header as the other DCS plugins (logo, name, workflow boxes, project box, Window and guide buttons); the workflow boxes are always shown. Without a project the start page gives the patch test field procedure; projects are created and opened from the project box. Needs DCS Core 1.3.0.
1.3.0	DCS Core's standard window (also for the Data View): floating the first time, docked only from the Window menu, and back as you left it after QGIS restarts. Lighter view backgrounds.
1.2.0	`.clraw`, SBET and LAS handling come from DCS Core, shared with CloudPro. Without a channel calibration in the capture, the measured default table is used. SBET files with a GPS week rollover read correctly.

2 Installing and opening AlignPro

AlignPro is one of three QGIS plugins that process a Canadian Lidar mobile mapper job one after the other. All three need the **DCS Core** plugin, which holds what they share: the job's processing project, the files each stage hands to the next, the look, the header and the window.

STAGE	PLUGIN	READS	FINALIZES
01	**PosPro**	The mapper captures (`.clraw` or `.anpp`), base station data	The trajectory: `trajectory.json` with SBET and SMRMSG
02	**AlignPro**	The captures and PosPro's trajectory	The Lidar mounting: `alignment.json`
03	**CloudPro**	The captures, the trajectory and the mounting	The point cloud: `cloud.json`, LAS 1.4, ground grid, report

Requirements

- QGIS 3.34 or later, including QGIS 4.
- DCS Core 1.15.0 or later. AlignPro 1.10.1 will not start with an older DCS Core: a message bar asks you to update it.

Install from the Canadian Lidar repository

1. In QGIS open `PLUGINS › MANAGE AND INSTALL PLUGINS › SETTINGS`, press **Add...** and enter the repository URL `https://canadian-lidar.github.io/qgis-plugins/plugins.xml`.
2. Press **Reload all Repositories**, then install **DCS Core** first and **AlignPro** after it.
3. Restart QGIS. Later versions appear under `UPGRADEABLE` in the same window; update DCS Core together with the plugins.

Without internet access, use `INSTALL FROM ZIP` in the same window with `dcs_core-1.15.0.zip` first, then the AlignPro zip.

Opening AlignPro

- `PLUGINS › CANADIAN LIDAR › DATA SYSTEMS › ALIGNPRO`, or the AlignPro button on the Plugins toolbar.
- From another DCS plugin: click the **AlignPro** box in its header to open AlignPro on the same project.
- This guide: the `?` button in the header, `USER GUIDE (THIS VERSION)` in the project box, or `HELP › PLUGINS › ALIGNPRO USER GUIDE`. The guide that opens always matches the installed version.

3 The AlignPro window

PosPro, AlignPro, CloudPro and RemixPro share the same header and the same kind of window, so what you learn here applies to all of them.



Figure 3.1 The header. Left: the logo, the plugin name and version, then two rows of breadcrumbs: the chain of stages and this plugin's own steps. Then the project box, and the user guide button at the right.

The header

PART	WHAT IT DOES
Plugin name and version `PosPro v2.1.0`	The plugin and the version you are running. Quote it when you report a problem; the user guide (`?`) is the one for this version.
The chain `PosPro › AlignPro › CloudPro`	The state of each stage on the open project, by colour: **green** finalized and current, **amber** out of date (an earlier stage was run again, the raw data or a shared setting changed), **blue** running, **red** failed, **grey** not run yet. The plugin you are in is underlined. Hover a name for the state and the reasons; click another stage to open that plugin on the same project. The chain refreshes every few seconds.
AlignPro's steps `PREPARE › CALIBRATE › REPORT`	The step list on the left in three phases: Prepare (Data, Installation, Quality), Calibrate (Pairs, Calibration: Apply finalizes the result for CloudPro) and Report (Analyze, Report). Each step is coloured the same way (green done, amber needs a check, red needs attention, blue running, grey not done yet); the current one is underlined. Hover a step for what is done and what to do next; click it to go there.
Project box	The open project's name; click it for the project menu (Projects), which also opens the Data View. It turns red while AlignPro is working.
`?`	Opens this guide.

Moving through the steps

Under the step pages, `‹ BACK` goes to the previous step and `NEXT: <STEP> ›` to the next one; Next is greyed out until a required step is done and reads `SKIP TO <STEP> ›` on optional steps. Hover either button for what is done and what to do. On the last step Next becomes `CONTINUE IN CLOUDPRO ›`, which opens CloudPro on the same project once the calibration is finalized. The step list on the left and the steps in the header still open any step at any time.

The window

AlignPro opens as a **window of its own**, centred on the QGIS screen the first time. Use the system's title bar to move it, maximise, minimise or close it; it does not dock into the QGIS window. AlignPro remembers its place and size, or that it was maximised, and opens the same way next time, also after QGIS restarts. `RESET WINDOW` in the project menu brings it back to its default size, centred.

4 Projects and the job folder

PosPro, AlignPro and CloudPro work on one shared **processing project** per job. The mapper's capture folders are never changed: everything the plugins make goes in a `Processing` folder.

PATH	HOLDS
`Job_Oct20/`	The job folder: the capture folders recorded by the mapper (read only)
`Job_Oct20/Processing/<Name>/<Name>.dcsproj`	The shared project: captures, shared settings (coordinate system, base station, antennas, Lidar mounting, vehicle) and the state of each stage. A `.bak` copy is kept beside it.
`01_PosPro/`, `02_AlignPro/`, `03_CloudPro/`	Each stage's own settings file, `work/`, `log/`, and `out/` with what it finalizes
`<Name>.qgz`	The QGIS project, when you save it from the project menu

Mapper 2.10 projects

From mapper 2.10 the mapper keeps its own projects. Use the mapper project's folder as the job folder:

PATH	HOLDS
`Deck_Test/project.json`	The mapper project: name, id, next line, operator, vehicle
`Deck_Test/lines/Deck_Test_L001_20261104_124512/`	One survey line (one Start/Stop), its files named `Deck_Test_L001_20261104_124512_0001.clraw`, `_0002` … Captures made before 2.10 that the mapper grouped into the project keep their old names here.
`Deck_Test/nav/Deck_Test_NAV_20261104_120000_0001.clraw`	A navigation recording: everything the Certus sent while the project was open (no Lidar). A new one starts at each power-on, project opening, UTC date change and every 2 GB.

If you pick the project's `lines` or `nav` folder, or a line folder, the New project dialog takes the mapper project's folder as the job folder. The survey lines are recognised from what the files record (their line number and capture), not from their names, so a line split over several files is one line and old and new captures mix freely, in recording order. Navigation recordings are never taken as survey lines.

The Processing folder can also live on another drive (choose **Elsewhere…** when you create the project). Paths are stored relative to the job folder where possible, so a job can be copied or moved as a whole.

The project menu

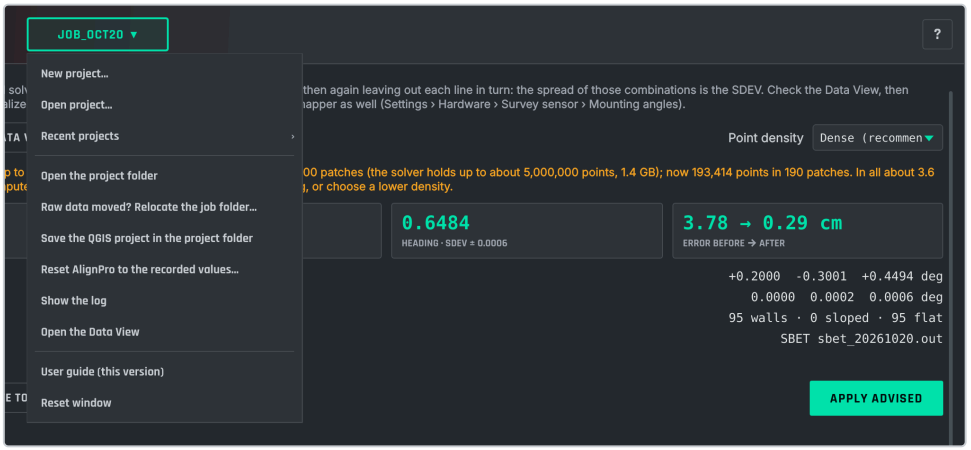


Figure 4.1 The project box menu in AlignPro.

ITEM	WHAT IT DOES
NEW PROJECT…	Creates a shared processing project in a job folder (below) and opens it.
OPEN PROJECT…	Opens a `.dcsproj`, for example the one PosPro made. The file dialog also offers older stand-alone AlignPro projects (`*.alignpro.json`); they open without the workflow boxes and cannot finalize.
RECENT PROJECTS	The last 15 projects AlignPro opened, as *name — folder*; *(none)* when the list is empty. Projects that no longer exist are left out. When AlignPro starts it reopens the most recent one.
OPEN THE PROJECT FOLDER	Opens `Processing/<Name>/` in your file manager.
RAW DATA MOVED? RELOCATE THE JOB FOLDER…	Asks for the job folder's new location and finds the captures there (see below).
SAVE THE QGIS PROJECT IN THE PROJECT FOLDER	Saves the QGIS project, with AlignPro's map layers, as `<Name>.qgz` in the project folder. If another QGIS project is open, AlignPro asks before saving it under the new name.
RESET ALIGNPRO TO THE RECORDED VALUES…	After a confirmation: puts the mounting angles, lever arm and latency back to the values recorded in the `.clraw` files, uses every line again, removes the calibration result and places the inspection boxes again. The processing settings stay (Settings has *Restore defaults*), and nothing is finalized by the reset itself. Available once the data is read.
SHOW THE LOG	The *AlignPro log* window: this session's messages followed by the project's own log (the last 500 entries).
USER GUIDE (THIS VERSION)	Opens this guide.
OPEN THE DATA VIEW	Opens the Data View window (Data View).
RESET WINDOW	The window back to its default size, centred.

Creating a project

1. Choose NEW PROJECT… in the project box.
2. **Job folder**: the folder that holds the mapper's capture folders. The project name defaults to the date of the first capture with a number, for example *2026-11-03_1*; when that project exists already the next number is offered

(*2026-11-03_2*), so a second processing of the same job (another base station, other settings) gets its own folder. Type any other name if you prefer.

3. **Processing folder**: inside the job folder by default; **Elsewhere…** to put it on another drive.
4. Press OK. The project is created and opened. AlignPro then looks for the survey lines' `.clraw` files in the job folder (mapper 2.10 navigation recordings are left out); when the project has no captures recorded yet it lists the files it found and asks whether to use them, then reads them straight away (see the Data chapter).

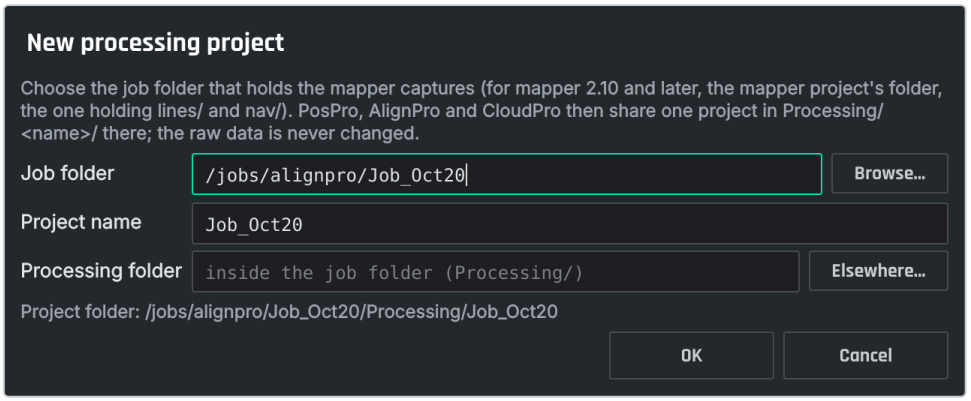


Figure 4.2 The New DCS project dialog (the same in every plugin).

A project made in any of the three plugins is the same project: open the `.dcsproj` in another plugin, or click its box in the header.

When the raw data has moved

If the captures are not where they were (another drive letter, a copied job), AlignPro says so when the project opens and asks for the job folder's new location; `RAW DATA MOVED? RELOCATE THE JOB FOLDER…` in the project menu does the same at any time. The captures are found again by their place in the job folder; nothing goes out of date because of the move.

Out-of-date results

Each stage records what it was built from. When an earlier stage finalizes again, the raw data changes, or a shared setting it depends on is edited, the stage turns **amber (out of date)** in the chain in the header, with the reason on hover. Run that stage again to bring it up to date. A run that fails or is cancelled keeps the earlier results.

5 Working in AlignPro

Below the header, AlignPro has the seven workflow steps on the left and the page of the current step on the right. Work down the steps; you can go back to any step at any time.

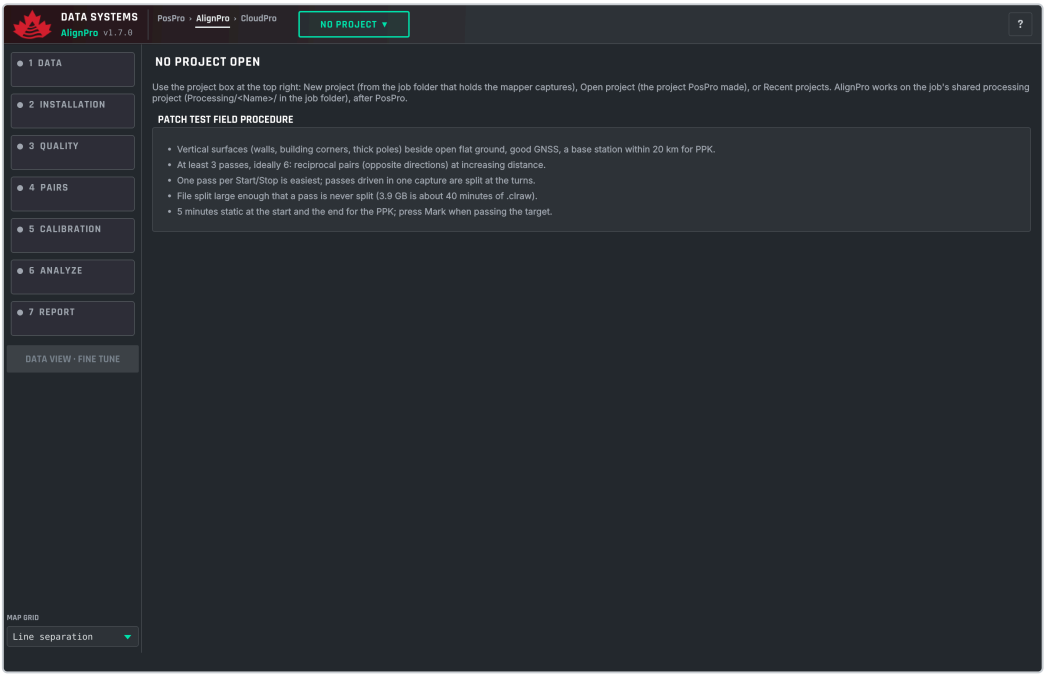


Figure 5.1 AlignPro without a project: the start page gives the patch test field procedure. The steps are greyed out until a project is open.

The start page

With no project open, AlignPro shows `NO PROJECT OPEN` : where projects come from (the project box) and the patch test field procedure (Chapter The calibration drive). AlignPro reopens the project you used last when it starts, so you usually see this page only the first time.

The steps

Each step is a pill with its number, its name and a second line that says what it holds. The colour gives its state:

STEP	SECOND LINE	GREEN WHEN	AMBER OR RED WHEN
1 Data	lines and total driving time	the files are read without warnings	amber: a data warning (Chapter Troubleshooting)
2 Installation	the lever arm	a lever arm is set	amber: the lever arm is 0, 0, 0
3 Quality	worst PPS lock and lost packets	PPS lock 100 % and no lost packets	amber: PPS lock lost or packets lost
4 Pairs	pairs and patches	at least 2 reciprocal pairs and 20 wall patches	amber: fewer, but at least 6 patches; red: fewer than 6 patches
5 Calibration	the advised angles, or what to do next	calibrated, applied and finalized, no warnings	amber: a calibration warning, *apply to finalize*, or *out of date: re-run*
6 Analyze	latency, lever, angles	optional; always grey	

7 Report	the last report's file name	a report was written in this session	–

Only **1 Data** is available until the captures have been read; the other steps follow.

Under the steps

- `2 CHECK / FINE TUNE…` opens the Data View as a floating window (Chapter Data View and fine tuning).
- The live figures: **AVG ERROR**, the mean point-to-plane distance over the planar patches with the current angles (green below 1 cm, amber below 2.5 cm, red above); **SDEV** and **SEP**, the average standard deviation and line separation of the map grid; the trajectory in use (*POSPRO*, *SBET FILE* or *REAL-TIME*) and *LOCAL ENU*, the local frame AlignPro works in; and the current roll (R), pitch (P) and heading (H). Hover for an explanation.
- `MAP GRID` : which value the grid layer on the QGIS map shows: *Mean height* (with a hillshade), *Standard deviation*, *Line separation* (the default), *Count* or *Lines per cell*. See Chapter Map layers.

Messages and running jobs

Reading, calibrating, applying, analysing and exporting run as QGIS background tasks. While one runs, the project box turns red, the status line at the bottom shows what is being done, a progress bar and a `STOP` button appear, and the pages cannot be changed. `STOP` cancels the job; a stopped or failed job keeps the earlier results. Starting a second job meanwhile gives *Wait for the current job to finish (or stop it).*

Short results appear in a bar above the status line. Warnings and errors also appear in the QGIS message bar, and every message goes to the log (`SHOW THE LOG` in the project menu).

6 The calibration drive

A good calibration depends more on the drive than on the software. Plan it with the crew; they record it like any other job.

Site

- Vertical surfaces (building walls, building corners, thick poles) beside open, flat ground. Walls drive the heading; flat ground and walls together drive roll and pitch. Walls facing the road and walls across it are both useful.
- Good GNSS sky view and a base station within 20 km, so PosPro can make a fixed post-processed (PPK) trajectory.

Passes

- At least 3 passes, ideally 6, as reciprocal pairs (out and back in opposite directions) at increasing distance from the target. With three or more passes AlignPro can leave each one out in turn and give a real SDEV; with two it reports the solver's formal standard deviation instead.
- Drive steadily. One pass per Start/Stop is easiest. Several passes in one capture also work: AlignPro splits them at the turns and drops the stops (Settings, *Passes and pairs*).
- Set the mapper's file split large enough that a pass is never split across files (3.9 GB is about 40 minutes of `.clraw`).
- Stay static for 5 minutes at the start and at the end for the PPK, and press **Mark** on the mapper when passing the target.

Before and during the drive

- Enter the measured Lidar lever arm in the mapper. AlignPro solves the angles, not the lever arm.
- Check that the Lidar time lock (PPS) stays locked. A pass with a lost lock or lost packets is best left out of the calibration (Chapter Quality).

Processing order

Process the job in PosPro first, then open the same project in AlignPro. AlignPro can start on the mapper's real-time trajectory or on another post-processed SBET file for a quick look, but the final calibration should use PosPro's trajectory.

7 1 Data

The Data page shows which captures and which trajectory the calibration uses, what the files contain, and the passes (**lines**) AlignPro made of them.

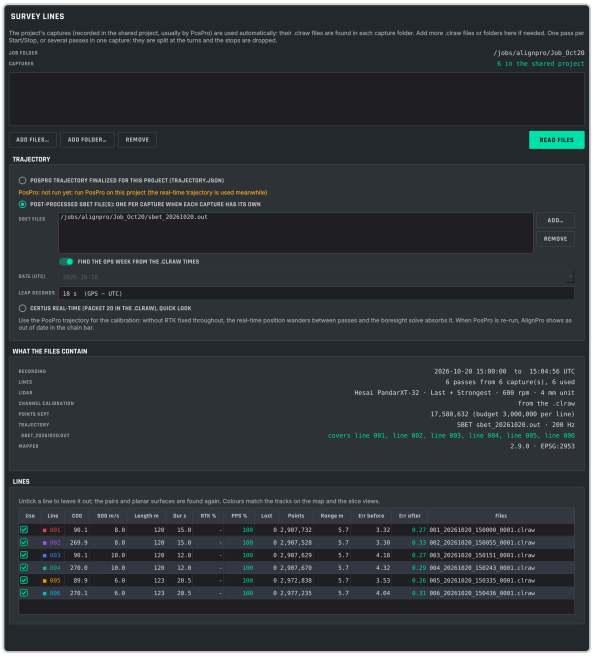


Figure 7.1 The Data page after reading two captures with an SBET file as trajectory. PosPro has not run on this project, so its trajectory is not available.

Captures

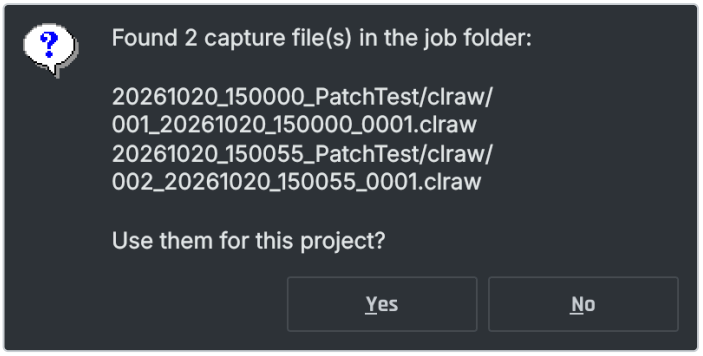


Figure 7.2 After **NEW PROJECT…** : AlignPro lists the captures it found in the job folder. Yes records them as the project's captures and reads them.

Job folder and **Captures** come from the shared project: AlignPro uses the captures recorded there (usually by PosPro) and finds the `.clraw` files in each capture folder. When the project has no captures recorded, AlignPro searches the job folder (*the job folder is searched*); when AlignPro is the first stage on the job, **READ FILES** records the files it uses as the project's captures. If raw files are missing, a red **Missing** line points to **RELOCATE THE JOB FOLDER**. Files inside the `Processing` folder are never read as raw data.

The list under it holds extra `.clraw` files or folders for this calibration only:

- **ADD FILES…** adds `.clraw` files; **ADD FOLDER…** adds a capture or export folder (its `.clraw` files are found inside); **REMOVE** takes the selected entries off the list.

- **READ FILES** saves the list and the trajectory choice and reads everything again. It also places the inspection boxes again.

The first read decodes every capture and keeps a cache in `02_AlignPro/work/`; later opens read the cache and are much faster. Packets recorded while the vehicle is slower than the minimum pass speed are skipped while reading (they would only repeat the same spot and use up the point budget). A capture with turns and stops is split into straight passes, named after the capture's line number with a letter (`003a`, `003b`); a capture that is one straight pass keeps its number (`001`).

Trajectory

The trajectory gives the vehicle's position and attitude for every point. Choose one of three:

OPTION	USE IT
POSPRO TRAJECTORY FINALIZED FOR THIS PROJECT (TRAJECTORY.JSON)	For the final calibration. This is the choice for a new project. The line under it says PosPro's state: *finalized and current* (green), *finalized, but out of date* with the reasons, *PosPro is processing now*, *the last PosPro run failed*, or *not run yet* (amber). While PosPro has not finalized a trajectory, AlignPro uses the Certus real-time trajectory and says so in a warning. GPS week and leap seconds come from PosPro's file.
POST-PROCESSED SBET FILE(S)	One SBET for the whole drive, or **one per capture** (see below). **ADD…** picks one or more files (`*.out`, `*.sbet`) and selects this option; **REMOVE** takes the selected ones out. A date in the first file's name such as `sbet_20261020.out` fills the date. **FIND THE GPS WEEK FROM THE .CLRAW TIMES** (on by default) works out the GPS week from the capture times; turn it off to give the **DATE (UTC)** yourself. **LEAP SECONDS** is GPS − UTC (18 s since 2017). Without a file name this option falls back to the real-time trajectory.
CERTUS REAL-TIME (PACKET 20 IN THE .CLRAW), QUICK LOOK	A first look in the field. Without RTK fixed throughout, the real-time position wanders between passes and the solved angles absorb it; AlignPro warns when it is RTK fixed less than 95 % of the time.

One SBET per capture. With the older MapPro format the Certus was logged only during each capture, and every Start/Stop made a new project folder, so each capture has its own SBET. Add every capture folder to the project (**ADD FOLDER…** above) and every SBET to the list. AlignPro places each SBET by its GPS time and uses it for the captures it covers; the order of the list does not matter. Nothing is interpolated between two SBETs. Under *What the files contain*, each SBET is listed with the lines it covers, and a warning names any line that no SBET covers (that line is then left without points).

A calibration solved on anything other than PosPro's trajectory is marked on the Calibration page (*Solved on the SBET file trajectory, not PosPro's…*) and in `alignment.json`. When PosPro later finalizes, AlignPro turns stale in the workflow boxes.

What the files contain

ROW	SHOWS
Recording	First and last time of the lines (UTC)
Lines	Passes, the captures they come from, and how many are used
Lidar	Model, return mode, rotation speed and range unit
Channel calibration	*from the .clraw*, or *missing: nominal angles used* (amber)
Points kept	Points read in all lines and the budget per line (Settings, *Points kept per line*)
Trajectory	The trajectory in use and its rate
RTK fixed	The worst line's share of RTK fixed epochs, when the trajectory carries a fix type
Mapper	Mapper software version and coordinate system recorded in the capture

Data warnings appear in amber under the rows (see Chapter Troubleshooting).

Lines

One row per pass. The line colour is the same on the map, in the Data View and on the other pages.

COLUMN	MEANING
Use	Untick to leave the line out. AlignPro saves the choice and finds the pairs and planar surfaces again; calibrate again afterwards.
Line	Name in its colour. A tooltip says when the trajectory does not cover the whole pass.
COG, SOG m/s	Course over ground (degrees) and speed
Length m, Dur s	Length and duration of the pass
RTK %	Share of RTK fixed epochs (green from 99 %, amber above 80 %, red below); - when the trajectory has no fix type, as with an SBET file
PPS %	Share of Lidar status packets with the PPS time lock (green at 100 %)
Lost	Lost Lidar packets (red when any)
Points, Range m	Points kept and their median range
Err before, Err after	The line's mean point-to-plane error (cm) before and after the last auto calibration
Files	The `.clraw` files of the line

8 2 Installation

The Installation page holds the lever arm, mounting angles and latency the calibration starts from. They come from the hardware record in the first capture (shared in the project).

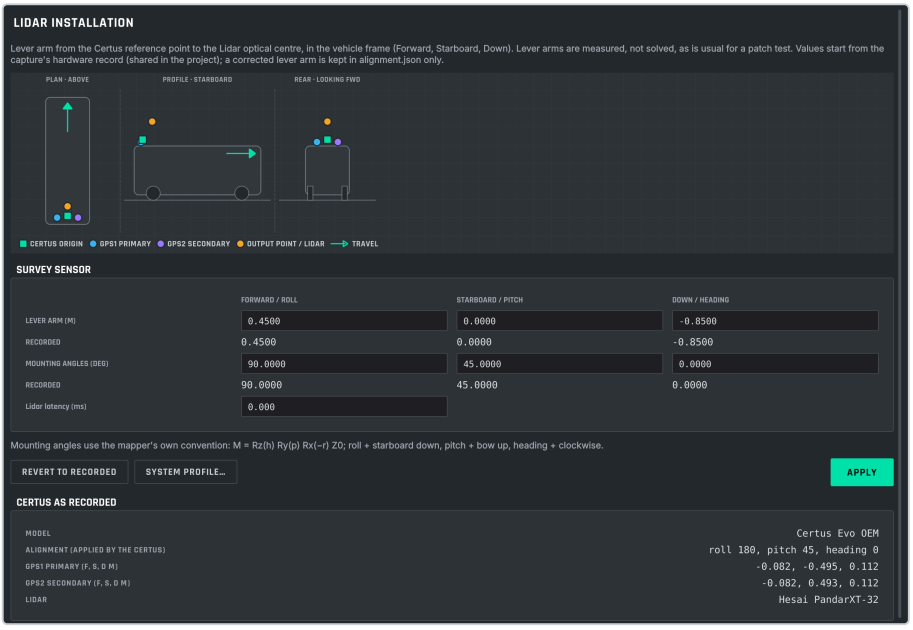


Figure 8.1 The Installation page with the vehicle drawing, the survey sensor values and the Certus as recorded.

- The **drawing** shows the vehicle from above, from starboard and from behind, with the Certus origin, both GNSS antennas and the Lidar (output point) at the lever arm you type. It is the mapper's own mounting drawing and follows the vehicle type in the capture.
- `LEVER ARM (M)`: from the Certus reference point to the Lidar optical centre, **Forward**, **Starboard**, **Down**. It is measured, not solved. The value recorded in the capture is shown under each field. A lever arm of 0, 0, 0 turns the step amber: enter the measured one.
- `MOUNTING ANGLES (DEG)`: roll, pitch and heading in the mapper's convention (Chapter Reference), with the recorded values under them.
- `LIDAR LATENCY (MS)`: added to the Lidar time. Leave it at 0 while the PPS time lock is good.
- `REVERT TO RECORDED` puts the recorded lever arm and angles back in the fields and sets the latency to 0. Nothing changes until you press Apply.
- `SYSTEM PROFILE…` saves the applied values as the profile of this system, or uses a saved profile (below).
- `APPLY` re-georeferences every line with the values in the fields, finds the planar surfaces again, and finalizes `alignment.json` for CloudPro.
- `CERTUS AS RECORDED`: the Certus model, the alignment the Certus applies, both antenna offsets and the Lidar model, for reference.

SYSTEM PROFILES

A system profile keeps what is true of the hardware rather than of one job – the Lidar lever arm, mounting angles and latency here, the antenna offsets, rear axle, IMU and vehicle in PosPro – so once you have checked them (a patch test here, `CHECK ANTENNAS` in PosPro) the next job from the same system starts with them. Profiles are kept on this computer and can be exported to a file and imported on another.

- A profile knows the system it was made for: the mapper unit's serial number (from the survey line's `session.json`) and the Certus serial number (from `hardware.json`).

- **Save:** after `3 APPLY ADVISED & FINALIZE`, press `SAVE TO SYSTEM PROFILE…` (Calibration) or `SYSTEM PROFILE…` (Installation). `NEW…` makes a profile for this job's system; `SAVE THIS JOB'S VALUES TO THE PROFILE` updates the selected one. Only the Lidar section is written; the antenna offsets PosPro saved stay as they are.
- **Use:** when the lines are read and a profile matches the system that recorded them, AlignPro shows the values that differ and asks whether to use them. `USE PROFILE VALUES` applies them as if you had typed them and pressed `APPLY`: the lines are georeferenced again, the profile's mounting becomes the job's starting mounting (shared with CloudPro) and the result is finalized. You are asked once per job for each version of a profile.
- The table in the System profiles window shows the profile beside this job; values that differ are amber. A profile made for another system can still be used, after a question.
- Keep the profile up to date when the hardware changes (the Lidar is remounted, an antenna is moved): save the new values after the next patch test.

> NOTE A corrected lever arm goes to `alignment.json` only. To make it permanent, enter it in the mapper as well. If the lines were recorded with different mountings or lever arms, AlignPro warns and uses the first line's values for all.

9 3 Quality

The Quality page checks the data before you trust a calibration. The mapper locks the Lidar clock to the GNSS (1PPS and GPRMC), so no latency should be needed: PPS lock should be 100 % and arrival minus sensor time about 1 ms.



Figure 9.1 The Quality page after calibration: the cards, and the sections Time sync and rates, Heading vs course and Residuals opened.

Cards

CARD	SHOWS
PPS lock	The worst line's PPS lock (green at 100 %, red below)
Lost Lidar packets	Lost packets in all lines (UDP sequence gaps)
RTK fixed	The worst line's RTK fixed share, or *SBET* when the trajectory has no fix type
Mean error now	The mean point-to-plane error with the current angles (green below 1 cm, amber below 2.5 cm)

Sections

Click a section's title to open or close it.

SECTION	WHAT TO LOOK FOR
Time sync and rates	Per line: Lidar packets per second (amber below 4,900; about 5,000 is normal), lost packets, PPS lock %, NMEA valid %, arrival − sensor time (amber above 5 ms), navigation rate (amber below 50 Hz), return mode, RPM and damaged chunks skipped. A line with a lost PPS lock or lost packets is a reason to leave it out (Data page, `USE`).
Positioning	Position standard deviation (3D) over each pass, and the fix type when the trajectory has one. Without accuracies (no SMRMSG with the SBET) the plot says *No position accuracies (SMRMSG missing)*.
Attitude accuracy	Roll, pitch and heading standard deviation over each pass; shown only when the trajectory carries them. Look for steady values without jumps.
Heading vs course	Per line: course, heading − course (crab), speed, length, duration. A land vehicle shows almost no crab; a steady offset with the same sign in both directions points at the trajectory heading, not the Lidar.
Residuals	The spread of point-to-plane residuals with the recorded angles (red) and the current ones (green), with their mean in the legend. After a good calibration the current curve is narrow and centred on 0.
Ranges	Points per range for each line, up to the range gate. Points should spread over the range gate, not all be close to the vehicle.

10 4 Pairs

The Pairs page shows which passes overlap and how many planar surfaces the calibration can use.

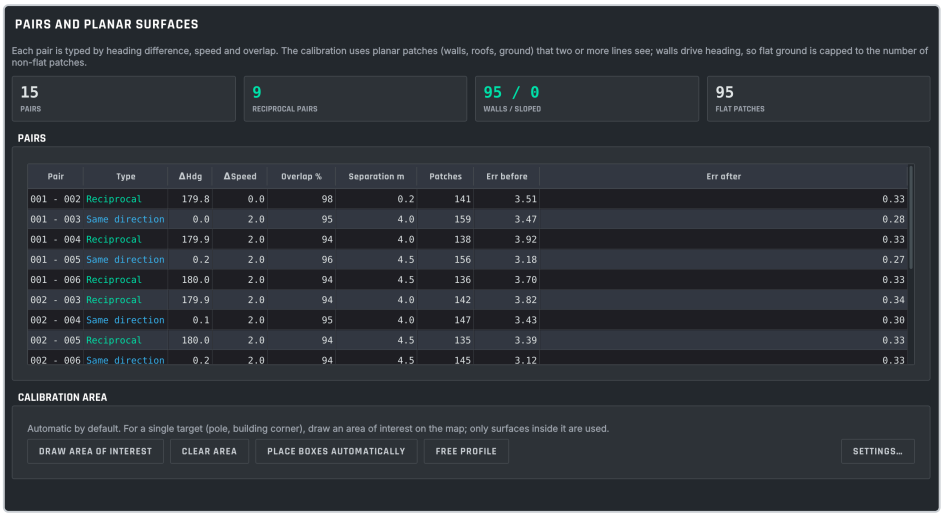


Figure 10.1 The Pairs page after calibration: one reciprocal pair with 142 shared patches.

Cards and the pairs table

The cards give the number of **Pairs**, **Reciprocal pairs** (green from 2), **Walls / sloped** patches (green from 20 walls) and **Flat patches**. A **patch** is a small flat piece of a surface (a voxel of *Patch size*, 2 m by default) that two or more lines see and that is flat within *Max thickness per line* in each of them. Walls drive the heading, so flat ground is capped to the number of walls and sloped patches (Settings, *Flat patches per non-flat patch*).

Each pair of lines is typed by heading difference, speed and overlap:

TYPE	MEANING (DEFAULT LIMITS)
Reciprocal	Opposite directions over the same ground (heading difference ≥ 150°): the strongest pair for roll and pitch
Same direction	Heading difference ≤ 30°, side by side
Crossing	Passes at an angle
Latency	Same direction, tracks less than 2 m apart, with clearly different speeds (≥ 2 m/s apart): shows a timing error
Additional	Too little overlap (below 20 %) or fewer than 3 shared patches

The columns are the pair, its type, ΔHdg (degrees), ΔSpeed (m/s), Overlap %, Separation m (distance between the two tracks), the shared Patches, and the pair's error before and after the last auto calibration (cm).

From the point cloud

A calibration error leaves a pattern in the finished points: between two passes driven in opposite directions the points of one pass sit a steady few centimetres from the other's, along the road or sideways, the other way round on the way back; a roll error makes the height difference grow with the distance from the drive line. CloudPro's *Check line separation* measures this on every line pair of the job, and this group shows what it found (it appears once CloudPro has checked):

COLUMN	WHAT IT IS

Driven	*both ways*, *same way* or *crossing*. Only lines driven both ways (or crossing) show a calibration error; driven the same way it cancels.
Along, Sideways, Up	The steady offset of the second line against the first, in cm, in the first line's direction of travel. Driven both ways, each pass carries half of it: *−8.0* along means each pass puts its points 4 cm ahead of (or behind) the vehicle.
Roll per 10 m	How much the height difference grows per 10 m from the drive lines.
Rest	What is left after the steady offset. A large rest is a trajectory error (PosPro), not the calibration.
Points at	*Calibration*, *Trajectory (PosPro)* or *Either*; the sentences under the table say why for the chosen pair.

What the patterns mean: an offset along the road that reverses with the driving direction is a timing latency, an along-track lever arm or pitch; sideways, a cross-track lever arm or heading; a growing height difference, roll. Calibrate again on that material: `USE THESE LINES AND THEIR OVERLAP` keeps only the chosen pair's passes and sets an area of interest 25 m each side of the stretch they share, then the pairs and surfaces are found again. Auto calibrate and Apply as usual; CloudPro's points go out of date, and after it updates them its next check should show the steady offset gone. The note above the table turns amber when CloudPro measured points made with an earlier calibration (or none).

From CloudPro, `OPEN IN ALIGNPRO` in the LINE SEPARATION tab opens this page on the same job with that pair selected. When CloudPro says *Along the road: not measurable here*, the walls facing along the road were too few or too scattered to give a number; the along column is then not a finding.

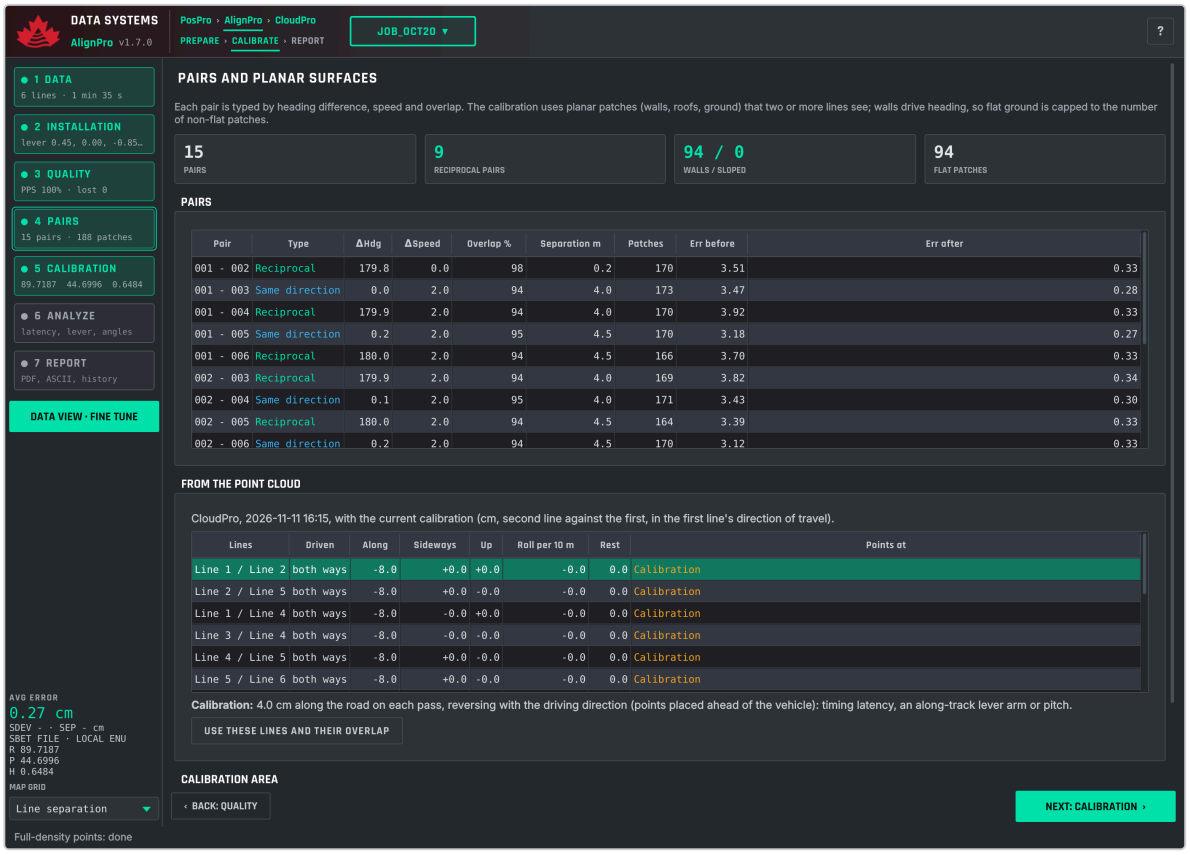


Figure 10.2 The Pairs page with CloudPro's findings on a job where every pass put its points 4 cm ahead of the vehicle: every pair driven both ways shows −8.0 cm along and points at the calibration.

Calibration area

BUTTON	WHAT IT DOES

DRAW AREA OF INTEREST	For a single target (a pole, a building corner). Click around the target on the QGIS map; right-click or double-click to finish (at least three points). Only patches inside the area are used, and the pairs and surfaces are found again. The area appears on the map as *Area of interest*.
CLEAR AREA	Back to the automatic choice over all the data.
PLACE BOXES AUTOMATICALLY	Puts the roll, pitch and heading inspection boxes on suitable surfaces again. The boxes are what the Data View shows under *Roll box*, *Pitch box* and *Heading box*, and the report's box pictures.
FREE PROFILE	Press on the QGIS map at the start of a profile, drag, and release at the end. A *AlignPro – Profile* window shows the points of every line along it. **CORRIDOR WIDTH** sets how wide it is; **PLAN VIEW** shows it from above instead of from the side.
SETTINGS…	The processing settings (below).

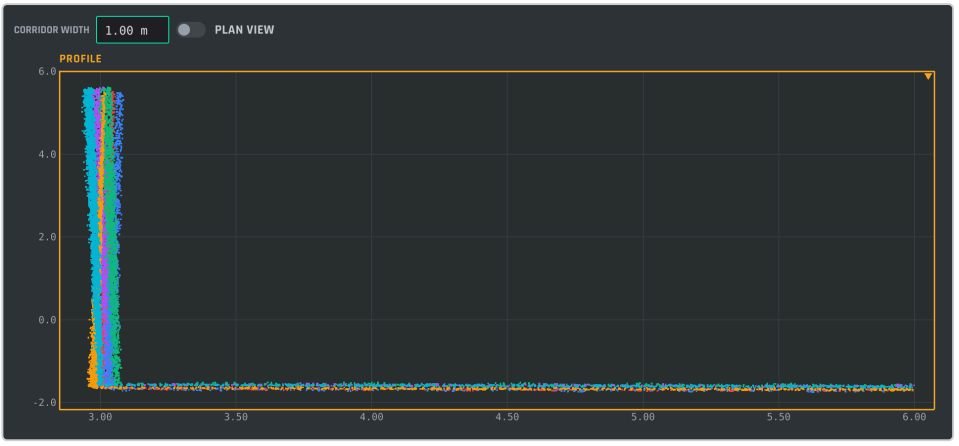


Figure 10.3 A free profile through a wall before calibration, at the recorded angles: the two passes (red and violet) show the wall a few centimetres apart.

Point density

The point density chooser beside **AUTO CALIBRATE** sets how many points the calibration and the Data View use. More points define each plane better and give steadier angles, at the cost of time and memory. The note under it lists the numbers in force and an estimate of the memory needed, as a share of this PC's memory; it turns amber when the job would use more than about 40 % and AlignPro refuses a density that would need more than about 80 %.

DENSITY	POINTS KEPT PER LINE	POINTS PER LINE IN EACH PATCH	PATCHES	USE
Standard	1,500,000	80	4,000	A quick look; what AlignPro used before 1.6.1.
Dense (default)	3,000,000	250	8,000	Calibration for delivery.
Very dense	6,000,000	600	15,000	Sparse scenes or long ranges; slow, needs a PC with plenty of memory.
Custom	The individual numbers in **SETTINGS…** (tab Data and Patches).			Your own balance.

Changing the points kept per line reads the lines again; the other numbers only rebuild the surfaces. Projects made before 1.6.1 keep their settings (Standard) until you choose another density.

Settings

The defaults suit the mapper; change them only for a reason. `RESTORE DEFAULTS` puts every value back. What happens on OK depends on what changed: data and pass settings read the lines again (and place the boxes again); the grid cell size only redraws the map grid; anything else finds the pairs and patches again.

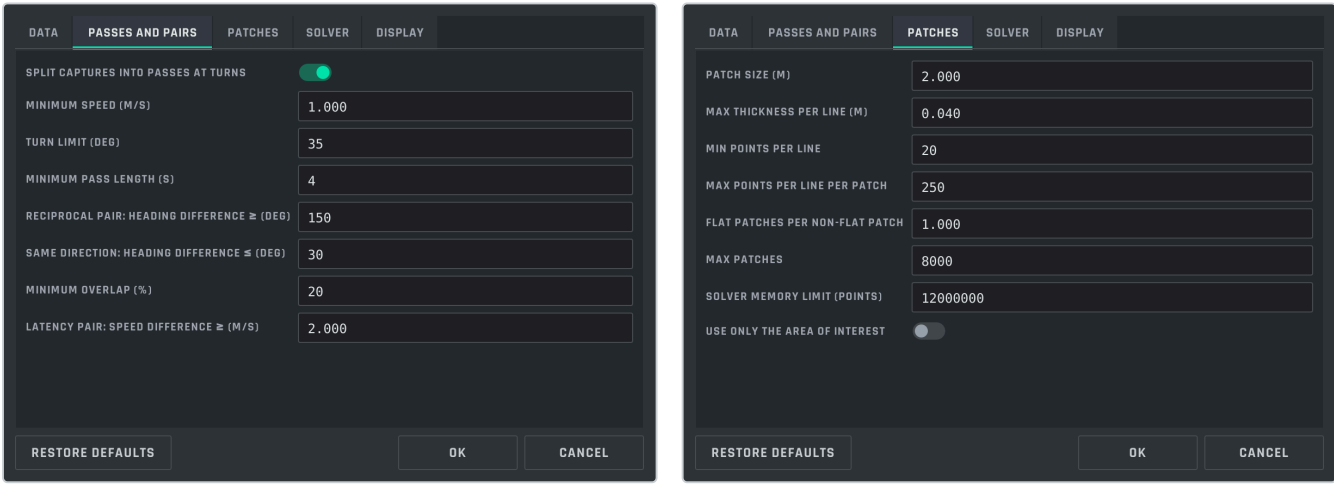


Figure 10.4 The Settings dialog, tabs Passes and pairs, and Patches. All tabs and values are listed in Chapter Reference.

11 5 Calibration

AUTO CALIBRATE solves roll, pitch and heading from every planar patch with all used lines, then again with each line left out when there are at least three. It starts from the current (applied) angles. It holds the AlignPro stage lock while it runs, so a second QGIS cannot work on AlignPro in the same project at the same time.

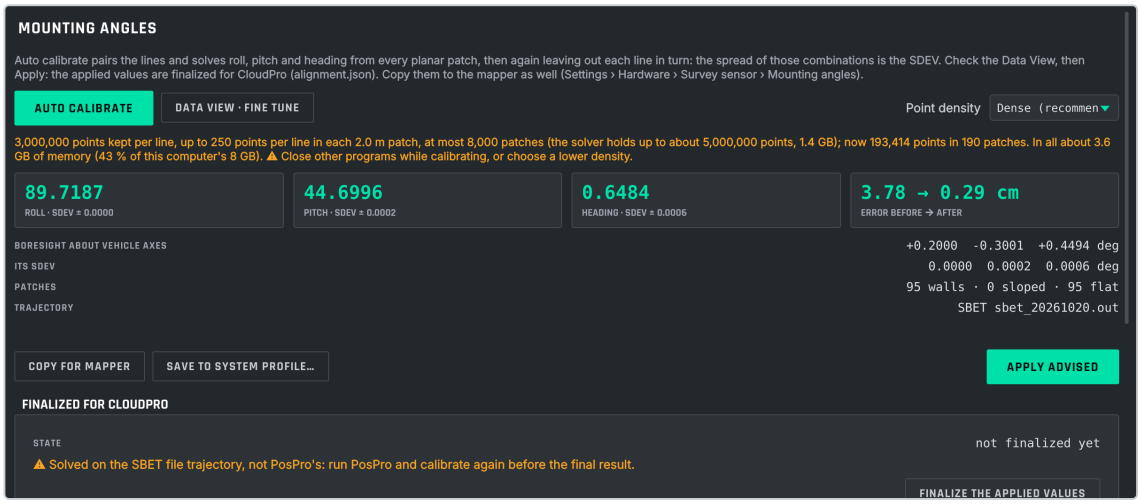


Figure 11.1 The Calibration page right after Auto calibrate: the advised angles with their SDEV, the error before and after, and *not finalized yet*.

Calibrate, check, apply

1. Press **AUTO CALIBRATE**. When it ends, the bar says *Calibration done … Apply to finalize it.* and the Data View opens with the advised angles as live values over the applied ones in grey.
2. Read the cards. Each angle card shows the advised value and its SDEV: green below 0.02°, amber from 0.02° to 0.05°, red above. **ERROR BEFORE → AFTER** is the mean point-to-plane error with the starting and the advised angles; after should be about 1 cm or less (green below 1 cm).
3. Check the Data View (Chapter Data View): the passes should fall on one surface in the roll, pitch and heading boxes.
4. Press **3 APPLY ADVISED & FINALIZE**. AlignPro re-georeferences every line with the advised angles and finalizes `alignment.json` for CloudPro.
5. Press **COPY FOR MAPPER** and paste the angles into the mapper: *Settings › Hardware › Survey sensor › Mounting angles*.

Under the cards

ROW	SHOWS
Boresight about vehicle axes	The correction from the starting angles, as small rotations about the vehicle's roll, pitch and heading axes. It stays readable near pitch 90°, where the mounting angles themselves can jump (Chapter Reference).
Its SDEV	The spread of that correction over the combinations
Patches	Walls, sloped and flat patches used
Trajectory	The trajectory the calibration was solved on

Calibration warnings appear in amber under these rows (Chapter Troubleshooting).

`COPY FOR MAPPER` copies the line `Roll 89.7187 Pitch 44.6997 Heading 0.6480` to the clipboard, to four decimals. It copies the **applied** angles, the ones finalized for CloudPro, so press it after `3 APPLY ADVISED & FINALIZE` (or after applying your own values). To copy trial values you have not applied, use `COPY FOR MAPPER` in the Data View.

Finalized for CloudPro

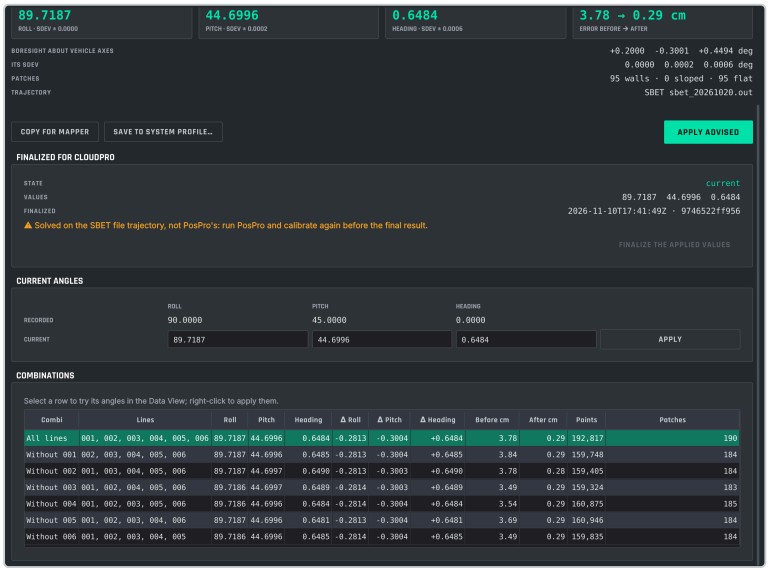


Figure 11.2 After Apply advised & finalize: the state is *current*, with the finalized values, time and identifier. The amber line is a reminder that this result was solved on an SBET file, not on PosPro's trajectory.

- **State**: *current*, *out of date* (with the reasons), *not finalized yet*, *running now* or *the last run failed*.
- **Values** and **Finalized**: the finalized roll, pitch and heading, the time (UTC) and the start of the publication's identifier.
- Amber notes: why it is out of date; *The applied values differ from the finalized ones*; or that the calibration was not solved on PosPro's trajectory.
- `FINALIZE FOR CLOUDPRO` (shown only when needed) writes `alignment.json` again with the values applied now. It is available when nothing is finalized, the publication is out of date, or the applied values differ from it, for example after `RESET ALIGNPRO TO THE RECORDED VALUES…` .

Current angles

Recorded shows the angles recorded in the capture; **Current** holds the applied ones. Type values and press `APPLY` to apply and finalize your own.

Combinations

One row per solve: *All lines*, then *Without 001*, *Without 002* and so on when there are at least three used lines. Columns: the lines used, the solved roll, pitch and heading, their change from the starting angles (Δ), the mean error before and after (cm), and the points and patches used. A solve that did not converge says so in red.

- **Select** a row to try its angles in the Data View (as trial values; nothing is applied).
- **Right-click** a row: `APPLY AND FINALIZE THESE ANGLES (…)` applies those angles and finalizes them for CloudPro; `COPY FOR MAPPER` copies them.

A combination clearly apart from the others points at a line that disagrees: check that line's quality and its error in the Data page, and leave it out if it is bad.

12 Data View and fine tuning

The Data View shows the real points in four views, plan, roll box, pitch box and heading box, and draws them again a moment after every change of an angle, the lever arm or the latency. Use it to check the automatic result and to adjust by hand when needed. The points are georeferenced with the live (trial) values; a grey **ghost** shows the same points with other values underneath.

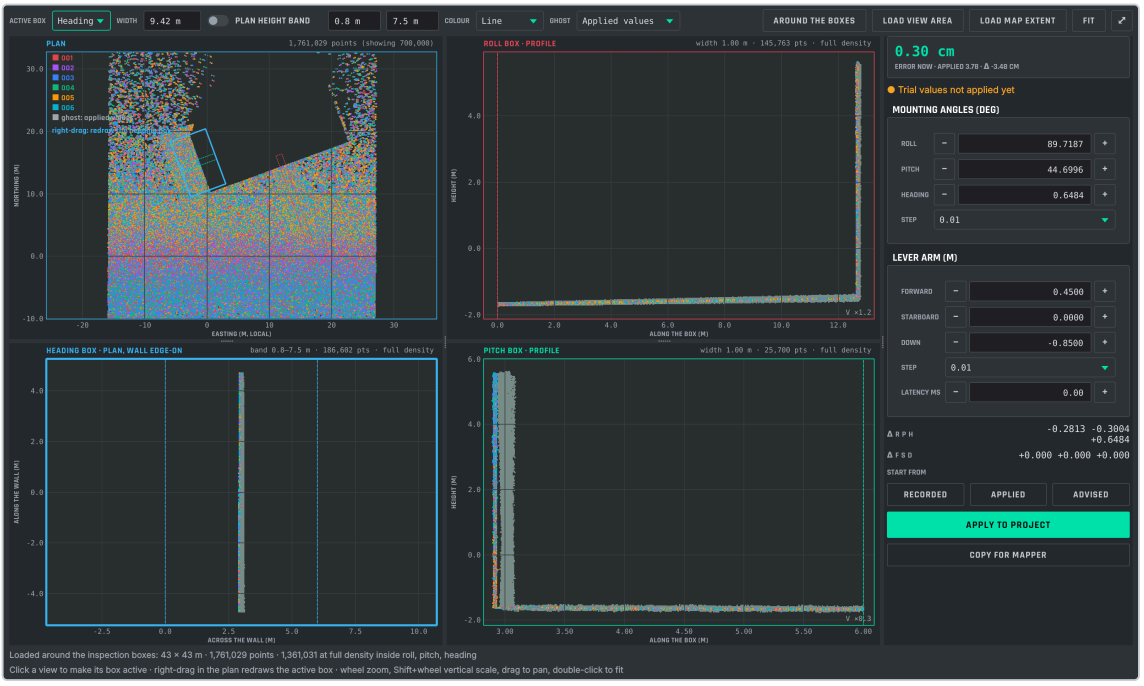


Figure 12.1 The Data View as it opens after Auto calibrate: plan, roll, pitch and heading views, with the advised angles as trial values and the applied ones as ghost. The panel on the right holds the values; the line under the views says what is loaded.

Opening it

- `2 CHECK / FINE TUNE…` under the steps or on the Calibration page: as a floating window.
- `OPEN THE DATA VIEW` in the project menu: where you left it.
- After reading the data and after Auto calibrate it opens by itself.

It is a standard DCS window: a window of its own (it does not dock), with a Window menu (top right of its toolbar) to maximise, reset or close it.

The four views

- **Plan** (top left): the loaded area from above, true to scale, with the line legend and the inspection boxes. The **active box** is drawn solid with an arrow at its end, the others dotted.
- **Roll box** (top right) and **Pitch box** (bottom right): the points along each box, height against distance.
- **Heading box** (bottom left): the wall seen from above, edge-on, within the box's height band. A heading error turns the passes against each other about a vertical axis, which a side view hardly shows; here it shows as passes turned apart.

A view without its box says so and how to draw one. In every view: the **wheel** zooms, **Shift + wheel** (or Ctrl + wheel) stretches the height, **drag** pans, **double-click** fits. The note at the top right of each view gives the point count, the width and whether the points are at full density.

Active box. Click a view (or pick it in **ACTIVE BOX**) to make its box the active one; its view gets a thicker frame. **Right-drag in the plan** redraws the active box, from where you press to where you let go, across the feature (through the wall for roll and pitch, across the wall for heading). **WIDTH** sets the active box's width; for the heading box it is the length of wall it covers. Boxes changed here are saved and shown on the map as well.

Which points are shown

The Data View loads a square area around the roll, pitch and heading boxes, and loads again whenever a box is drawn or moved, on the map (Pairs page) or in the Data View. Drawing an area of interest moves it there. The loaded area is outlined on the QGIS map as **Data View area**, and the line under the views says what is loaded.

Inside the boxes (and 1.5 m around them) you see **every point the Lidar recorded there**, not the thinned set the calibration reads. AlignPro re-reads the raw `.clraw` for just the seconds each pass was at a box. The first time takes a few seconds per box; the views show the thinned points meanwhile and fill in when the job ends. After that the points come from the project's `work/` folder. Outside the boxes the plan shows the thinned points, at most 700,000.

> NOTE **SETTINGS › FINE TUNE: POINTS PER INSPECTION BOX** (default 3,000,000) caps a very large box; above it the box's points are thinned at random and the line under the views says so. If the raw files cannot be found the views keep the thinned points.

The toolbar

CONTROL	WHAT IT DOES
ACTIVE BOX	*Roll*, *Pitch* or *Heading*: the box that a right-drag in the plan redraws and **WIDTH** sets
WIDTH	The active box's width (0.1 to 200 m)
PLAN HEIGHT BAND and its two heights	Shows only the points in this band above the ground in the plan, so walls and poles stand out. The heading box view always uses the box's own band.
COLOUR	*Line* (the line colours), *Height* or *Intensity*
GHOST	Grey points under the live ones, with: *None*, *Recorded angles* (the angles and lever arm recorded in the capture, latency 0), *Applied values* (the default), *Advised angles*, or *When view opened*. Coloured points are the trial values; grey shows where the points were.
AROUND THE BOXES	Loads the area around the inspection boxes again (after one of the two below)
LOAD VIEW AREA	Loads what the plan shows (a square of 10 to 300 m); boxes inside it at full density
LOAD MAP EXTENT	Loads what the QGIS map shows (up to 300 m across); boxes inside it at full density
FIT	Fits all four views to their points

The heading box

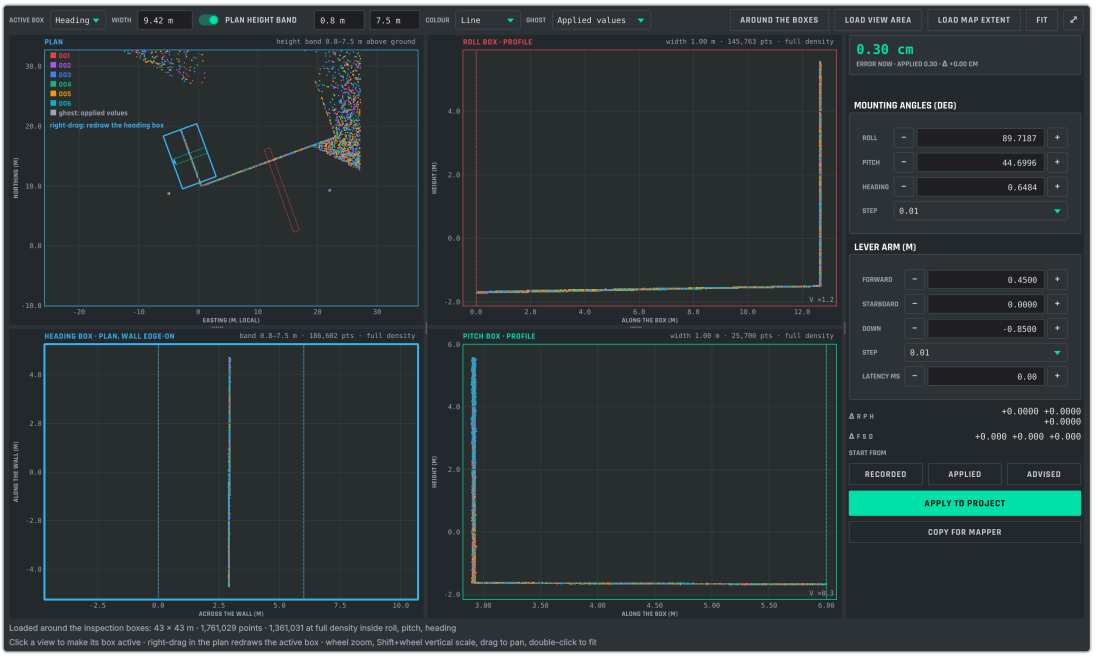


Figure 12.2 The heading box active (solid in the plan) with the plan height band on; bottom left, the wall seen from above, edge-on.

What a misalignment looks like

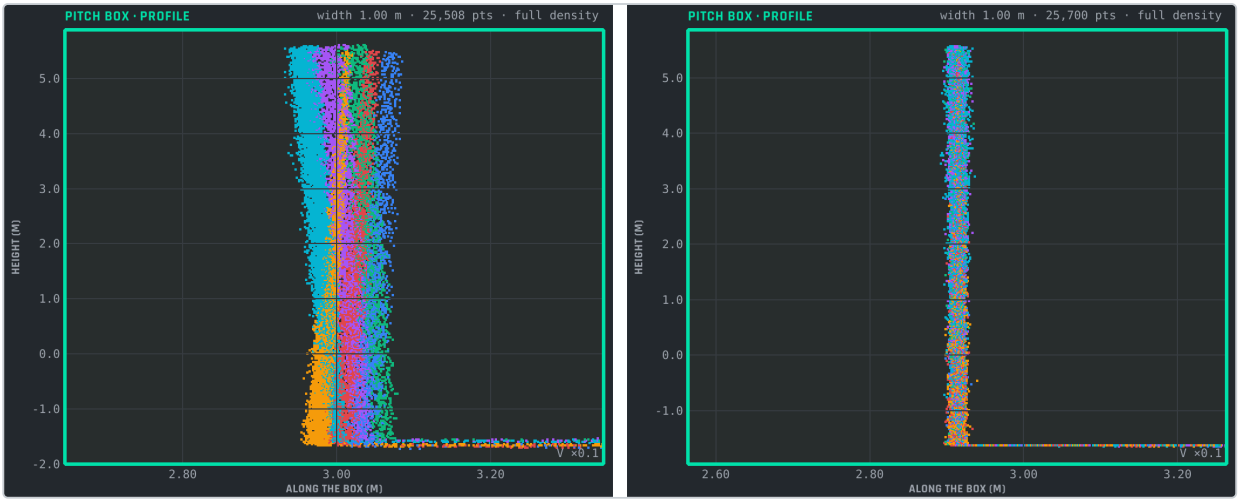


Figure 12.3 The same wall in the pitch box view, zoomed in, coloured by line. Left: with the recorded angles the two passes (red and violet) are a few centimetres apart and the wall looks thick. Right: with the advised angles they fall on one thin wall.

Adjusting by hand

The panel on the right holds the trial values:

- **Error now (planar patches)**: the mean error with the trial values; under it, with the applied values and the difference (Δ).
- `MOUNTING ANGLES (DEG)`: roll, pitch, heading. Each has `−` and `+` buttons (hold to repeat) and a field you can type in. `STEP` sets the step: 0.001 to 1°.
- `LEVER ARM (M)`: forward, starboard, down, with a step of 0.001 to 0.1 m, and `LATENCY MS` in steps of 1 ms. A latency change takes a little longer to draw.
- **Δ R P H, Δ F S D**: trial minus applied values.
- `START FROM`: `RECORDED` (the angles and lever arm recorded in the `.clraw`, latency 0), `APPLIED` (the values applied to the project) or `ADVISED` (the auto calibration result).
- `APPLY & FINALIZE THESE VALUES` applies the trial angles, lever arm and latency, finds the planar surfaces again and finalizes `alignment.json`.
- `COPY FOR MAPPER` copies the trial angles; the button reads *COPIED ✓* for two seconds.

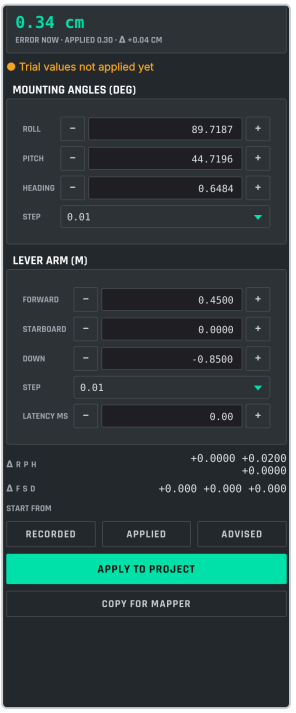


Figure 12.4 The panel with pitch raised by two steps of 0.01°: the error goes up by 0.03 cm.

Until you apply, the panel says ● **Trial values not applied yet**.

1. Start from `APPLIED` or `ADVISED`.
2. Watch the view that shows the angle you change: roll in the Roll box, pitch in the Pitch box, heading in the Heading box. Set `GHOST` to *Applied values* to see where you started.
3. Nudge until the passes fall on one surface and Error now is at its lowest.
4. Press `APPLY & FINALIZE THESE VALUES`, then copy the angles to the mapper.

> NOTE The fields show angles to 0.0001°, the lever arm to the millimetre and the latency to 0.1 ms. Values equal to the applied ones at that precision count as applied, so after `3 APPLY ADVISED & FINALIZE` the panel shows nothing to apply, and `APPLY` keeps the advised angles exactly (finalized as *calibration*).

13 6 Analyze

Analyze sweeps one value with everything else fixed and plots the mean point-to-plane error against it (the planar surfaces are fitted again at each step). The lowest point is the best value. Use it to check the latency or a lever arm component, or to confirm an angle.



Figure 13.1 A sweep of the mounting heading, ±0.3° in 21 steps, after applying the advised angles. The best value is the applied one.

1. Choose the **PARAMETER**: *Lidar latency (ms)*, *Lever forward (m)*, *Lever starboard (m)*, *Lever down (m)*, *Mounting roll (deg)*, *Mounting pitch (deg)* or *Mounting heading (deg)*. **Current value** shows the applied value; Centre, Range and Steps are filled with defaults (latency ±20 ms; lever ±0.3 m; roll and pitch ±0.3°; heading ±0.5°; 21 steps).
2. Adjust **CENTRE**, **RANGE ±** and **STEPS** (3 to 201) if needed, then press **ANALYZE**.
3. The plot shows the error at each step, with the best value as an amber line and the current value as a grey one. **BEST** gives the value and its error, and the Centre moves to it, so pressing Analyze again with a smaller range refines it.
4. **APPLY BEST VALUE** sets the parameter to the best value and finalizes `alignment.json`.

Latency shows as an along-track shift that flips with the driving direction, so it needs passes in both directions; a latency pair (Pairs page) helps. If the curve is jumpy, widen the range and add steps. Small values found on a quiet drive are usually noise: if it isn't broken, don't fix it.

14 7 Report

The Report page writes the calibration report as PDF and text, keeps the history of earlier calibrations, and exports corrected points.

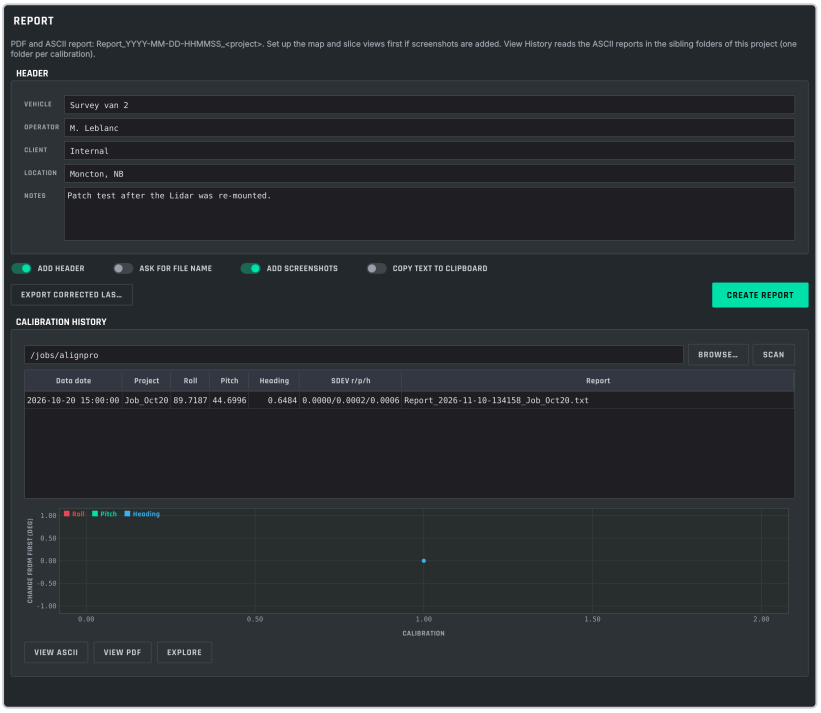
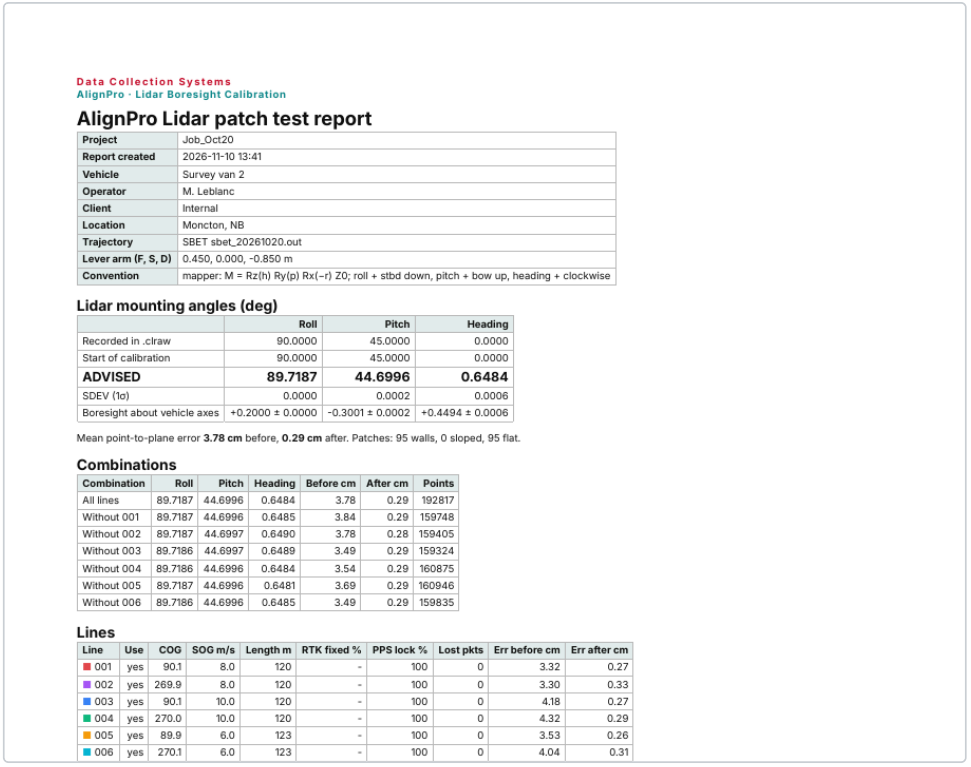


Figure 14.1 The Report page after writing a report. The history lists the reports found under the folder at the top.

Creating a report

1. Fill the **HEADER**: *Vehicle*, *Operator*, *Client*, *Location* and *Notes*. They are saved with the project.
2. Choose the options (remembered for next time):
   - **ADD HEADER**: the header fields go into the text report too.
   - **ASK FOR FILE NAME**: choose where to save; otherwise the report goes to `02_AlignPro/out/Report_YYYY-MM-DD-HHMMSS_<project>.pdf`.
   - **ADD SCREENSHOTS**: adds the QGIS map, the Data View's plan, roll, pitch and heading views, and the roll, pitch and heading boxes, each on its own page. Set up the map and the Data View first.
   - **COPY TEXT TO CLIPBOARD**: copies the text report.
3. Press **CREATE REPORT**. AlignPro writes the PDF and a `.txt` beside it, then asks whether to open the PDF.



Data Collection Systems
AlignPro · Lidar Boresight Calibration

AlignPro Lidar patch test report

Project	Job_Oct20
Report created	2026-11-10 13:41
Vehicle	Survey van 2
Operator	M. Leblanc
Client	Internal
Location	Moncton, NB
Trajectory	SBET sbet_20261020.out
Lever arm (F, S, D)	0.450, 0.000, -0.850 m
Convention	mapper: M = Rz(h) Ry(p) Rx(−r) Z0; roll + stbd down, pitch + bow up, heading + clockwise

Lidar mounting angles (deg)

	Roll	Pitch	Heading
Recorded in .clraw	90.0000	45.0000	0.0000
Start of calibration	90.0000	45.0000	0.0000
ADVISED	**89.7187**	**44.6996**	**0.6484**
SDEV (1σ)	0.0000	0.0002	0.0006
Boresight about vehicle axes	+0.2000 ± 0.0000	-0.3001 ± 0.0002	+0.4494 ± 0.0006

Mean point-to-plane error **3.78 cm** before, **0.29 cm** after. Patches: 95 walls, 0 sloped, 95 flat.

Combinations

Combination	Roll	Pitch	Heading	Before cm	After cm	Points
All lines	89.7187	44.6996	0.6484	3.78	0.29	192817
Without 001	89.7187	44.6996	0.6485	3.84	0.29	159748
Without 002	89.7187	44.6997	0.6490	3.78	0.28	159405
Without 003	89.7186	44.6997	0.6489	3.49	0.29	159324
Without 004	89.7186	44.6996	0.6484	3.54	0.29	160875
Without 005	89.7187	44.6996	0.6481	3.69	0.29	160946
Without 006	89.7186	44.6996	0.6485	3.49	0.29	159835

Lines

Line	Use	COG	SOG m/s	Length m	RTK fixed %	PPS lock %	Lost pkts	Err before cm	Err after cm
001	yes	90.1	8.0	120	-	100	0	3.32	0.27
002	yes	269.9	8.0	120	-	100	0	3.30	0.33
003	yes	90.1	10.0	120	-	100	0	4.18	0.27
004	yes	270.0	10.0	120	-	100	0	4.32	0.29
005	yes	89.9	6.0	123	-	100	0	3.53	0.26
006	yes	270.1	6.0	123	-	100	0	4.04	0.31

Figure 14.2 The top of the first page of the PDF report.

The report holds the project, trajectory, lever arm, latency and convention; the recorded, starting and advised angles with their SDEV and the boresight about the vehicle axes; the error before and after; the patches; every combination, line and pair; the warnings; the notes; and the files used. The text report ends with where to enter the advised angles in the mapper.

Calibration history

The history reads the text reports (`Report_*.txt`) found under the folder at the top, down to six folder levels, skipping `work` and `log` folders. It starts with the folder that holds this job's folder, so the reports of the other jobs beside it are listed too (each job's `Processing/<Name>/02_AlignPro/out`). Press **BROWSE…** to read another folder. **SCAN** reads the folder again.

- The table lists each report's data date, project, roll, pitch, heading, SDEV and file, oldest data first. Double-click a row to open its PDF.
- The plot shows the change of roll, pitch and heading from the first report, one point per report: a mounting that moves shows up as a trend.
- **VIEW ASCII** and **VIEW PDF** open the selected report; **EXPLORE** opens its folder.

Export corrected LAS

EXPORT CORRECTED LAS… writes the points of the used lines with the applied values to a LAS file (default `02_AlignPro/<Name>_corrected.las`), to check them in other tools. Coordinates are in the coordinate system recorded in the capture (EPSG:2953 when none is recorded) with ellipsoidal heights; each point keeps its intensity, its line number as point source ID and its GPS time. These are the thinned points AlignPro works with; full point cloud products come from CloudPro.

15 Finalizing for CloudPro and the mapper

CloudPro reads only what AlignPro finalizes; the mapper keeps its own settings, so copy the angles there too (**COPY FOR MAPPER**) and enter a corrected lever arm or latency by hand. Every apply finalizes: **3 APPLY ADVISED & FINALIZE**, **APPLY** on the Calibration or Installation page, a right-click **LOAD THESE ANGLES**, **APPLY & FINALIZE THESE VALUES** in the Data View, **APPLY BEST VALUE** on the Analyze page and **FINALIZE THE APPLIED VALUES**. Each one writes, in `02_AlignPro/out/`:

- `alignment.json`: the file CloudPro reads (schema `dcs.alignment/1`);
- `alignment_report.txt` (the text report) and `alignment_combinations.csv` (one row per combination).

Finalizing holds the AlignPro stage lock and writes a log in `02_AlignPro/log/`. If it fails, the earlier publication stays. CloudPro turns stale when AlignPro finalizes again.

What alignment.json holds

FIELD	CONTENT
`mount_deg`	The applied roll, pitch and heading, their SDEV (`sd`), and `source`: *calibration* when they are the advised angles, otherwise *manual adjustment*
`mount_recorded_deg`	The mounting the calibration started from, recorded in the capture, with its source
`lidar_lever_m`	The lever arm used (`value`), its `source` (*capture*, *measured* when you changed it, or *solved*) and the `recorded` one
`latency_s`	The Lidar latency applied, in seconds
`boresight_deg`	The correction relative to the mounting recorded in the capture, about the vehicle axes, with its SDEV
`quality`	The mean error now and with the recorded angles (cm), the SDEV, the lines used and their total, the patches, and the calibration warnings
`trajectory`	Which trajectory the values were solved on (*pospro*, *sbet* or *realtime*), PosPro's publication identifier, and a description
`convention`, `built_from`	The mounting convention (Chapter Reference), and what the result was made from, used to tell when it is out of date

An excerpt of the file finalized in this guide's example:

```
"mount_deg": {"roll": 89.71865, "pitch": 44.69973, "heading": 0.64803,
              "sd": {"roll": 0.00015, "pitch": 0.00061, "heading": 0.00079},
              "source": "calibration"},
"mount_recorded_deg": {"roll": 90.0, "pitch": 45.0, "heading": 0.0, "source": "capture"},
"latency_s": 0.0,
"trajectory": {"source": "sbet", "handover_id": null, "description": "SBET sbet_20261020.out"}
```

When AlignPro is out of date

AlignPro's box in the header turns ⚠ **stale** (reason on hover and on the Calibration page) when:

- PosPro finalizes a new trajectory, or finalizes one for the first time after AlignPro used an SBET file or the real-time trajectory;
- a raw capture file changes or is not found;
- the job's coordinate system or starting Lidar mounting (shared settings) is edited.

A stale result stays usable. To update it, run `AUTO CALIBRATE` again and apply.

16 Map layers and the command-line tool

Map layers in QGIS

AlignPro adds a layer group **AlignPro – <project>** to the QGIS project and zooms the map to the tracks when the data is read. The layers are drawn again whenever the lines, angles or boxes change. They use a local east-north frame centred on the job.

LAYER	SHOWS
Tracks	One line per pass in its line colour; a line left out is thin and dashed and marked *(not used)*. Attributes: line, use, COG, SOG, points.
Line starts	A circle at the start of each pass, so you can tell the driving direction.
Calibration boxes	The roll (red), pitch (green) and heading (blue) inspection boxes, each with a filled arrow at its end.
Planar patches	The patches used: walls (blue squares), sloped (amber diamonds), flat (grey circles), with the number of lines and the height. Hidden when added; tick it in the Layers panel.
Grid – <value>	A grid of the lowest surface per cell (cell size in Settings, *Display*, 0.5 m by default), coloured by the value chosen in **MAP GRID** under the steps: *Mean height*, *Standard deviation*, *Line separation*, *Count* or *Lines per cell*. Line separation is the quickest map check of a calibration: where passes overlap it should be close to zero everywhere.
Grid – hillshade	A hillshade over the grid, with *Mean height*.
Area of interest	The area drawn on the Pairs page; dashed when it is not in use.

The grid is a five-band GeoTIFF in `02_AlignPro/work/`. **SAVE THE QGIS PROJECT IN THE PROJECT FOLDER** keeps the map with the job.

Command-line tool

The plugin folder holds `tools/alignpro_cli.py`, which runs the same calibration without QGIS, for example on a processing server. It makes a stand-alone project (`<NAME>.alignpro.json`), not a shared DCS project, so it does not finalize `alignment.json`; it prints the results and writes the text report into PROJECT_DIR. It needs Python 3 with NumPy, and DCS Core beside the `alignpro` folder.

```
python3 alignpro_cli.py PROJECT_DIR NAME CLRAW_OR_FOLDER... [--sbet FILE [--date YYYY-MM-DD]]
                        [--apply] [--sweep PARAM CENTER HALF STEPS] [--set KEY VALUE]
```

ARGUMENT	MEANING
`PROJECT_DIR NAME CLRAW_OR_FOLDER…`	Where the project, cache and report go; the project name; the `.clraw` files or capture folders
`--sbet FILE`, `--date`	A post-processed SBET as trajectory (otherwise the real-time one), and its date when the GPS week should not come from the captures
`--apply`	Applies the advised angles to the project and prints the error with them
`--sweep`	A sweep as on the Analyze page; PARAM is `latency_ms`, `lever_f`, `lever_s`, `lever_d`, `roll`, `pitch` or `heading`
`--set KEY VALUE`	Overrides a setting, e.g. `--set voxel_m 1.5` (keys in Chapter Reference); repeatable

17 Troubleshooting

YOU SEE	WHY, AND WHAT TO DO
AlignPro needs the DCS Core plugin …	Install DCS Core, then restart QGIS.
No .clraw captures were found in … / No .clraw files found in the project	Check the job folder (relocate it if it moved), or add files on the Data page and press **READ FILES**.
Raw data not found when a project opens, or a red **Missing** line	Answer Yes and choose the job folder's new place, or use **RAW DATA MOVED? RELOCATE THE JOB FOLDER…**.
PosPro has not finalized a trajectory for this project yet: using the Certus real-time trajectory …	Run PosPro on the project, then read the data again in AlignPro and calibrate. Meanwhile choose an SBET file if you have one.
PosPro's trajectory is out of date … / Real-time trajectory is RTK fixed only …%	Re-run PosPro, then read and calibrate again in AlignPro.
The PosPro trajectory does not cover line … / SBET does not cover line …; check the SBET date and file.	Process that part in PosPro, or check the SBET date, the **FIND THE GPS WEEK** option and the leap seconds.
The Lidar lever arm recorded in the capture is 0, 0, 0 …	Enter the measured lever arm on the Installation page (and in the mapper), then Apply.
Line …: Lidar PPS lock lost in … status packets. / … damaged chunk(s) skipped.	Look at that line on the Quality page; untick it on the Data page if it disagrees.
Line … was recorded with a different Lidar mounting or lever arm …	The first line's values are used for all. Check each capture's mapper settings.
Not enough planar surfaces seen by two or more lines (…) / The calibration did not converge …	The passes do not share enough surfaces. Check the overlap on the Pairs page and the range gate in Settings; clear or widen the area of interest.
Few vertical surfaces (…): heading is weakly determined.	Drive past more walls, also walls across the road. Check the heading in the Heading box.
Combination spread above 0.05 deg …	One line disagrees with the others. Compare the combinations, the line errors and the Quality page; leave a bad line out and calibrate again.
The mounting pitch is near 90 deg …	Expected for the upright mounting. Enter all three angles exactly; judge the result on the boresight about vehicle axes and its SDEV (Chapter Reference).
The solved heading is close to minus the entered heading …	The heading sign of the mounting convention has not been confirmed for this case. Check the result in the Heading box before applying.
Wait for the current job to finish (or stop it).	Let it end or press **STOP**.
AlignPro is being processed by … on … since …	Another QGIS holds the AlignPro stage lock of this project. Wait until it finishes.
PDF not written (…). The ASCII report is at …	Close the PDF if another program has it open; check the folder is writable.

Limits in this version

- The lever arm is measured, not solved. Settings has a lever arm solve for diagnosis only.

- The inspection boxes are placed automatically (`PLACE BOXES AUTOMATICALLY`); for any other place use a custom profile in the Data View or `FREE PROFILE` .

18 Marine jobs: the sonar patch test

A job of sonar recordings (finalized in PosPro's Marine page) opens on one page, **Sonar patch test**, instead of the seven Lidar steps.

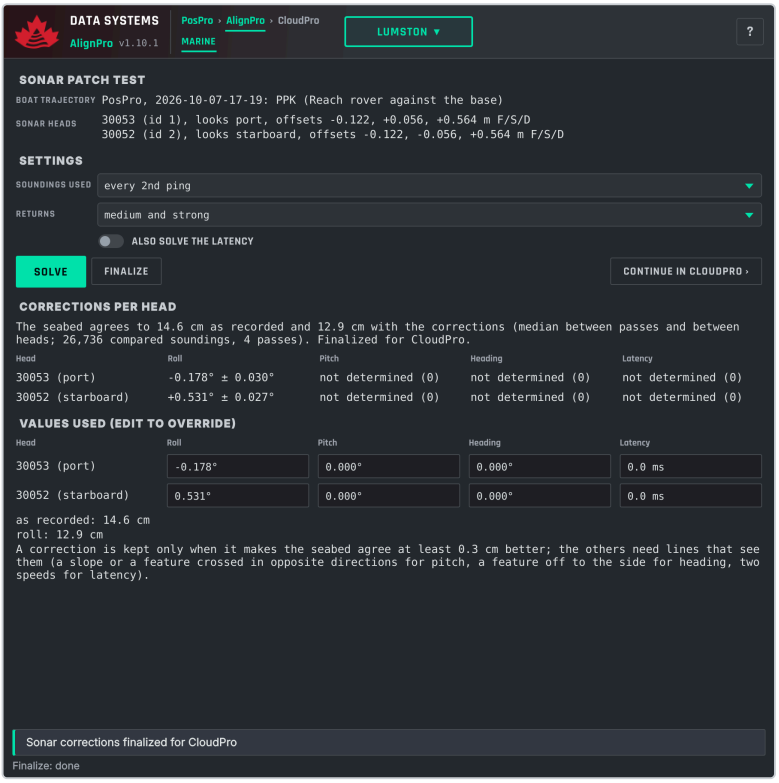


Figure 18.1 The Lumston survey after Solve and Finalize: the lines show a roll correction for each head; pitch, heading and latency are not determined by them.

How it works

The soundings are placed with PosPro's boat trajectory, each head's own attitude and its offsets. The track is cut into passes (the boat's straight runs). In 2 m cells the seabed of one head on one pass gives a plane, and the soundings of the other passes and of the other head in the same cell are compared with it. The corrections are solved one step at a time:

1. **Roll** for every head: it tilts the seabed one way on a pass and the other way on the reciprocal pass.
2. **Pitch and heading** per head: both move the seabed along the track, so each is kept only when it makes the seabed agree at least 0.3 cm better, and one is preferred when they cannot be told apart. They need a slope or a feature crossed in opposite directions (pitch) or a feature off to the side (heading).
3. **Latency** only with `ALSO SOLVE THE LATENCY`: it moves the seabed along the track as the pitch does, and is told apart only with lines run at different speeds.

The comparisons between passes of one head and between the two heads count equally, although there are far fewer of the second kind.

Settings

SETTING	MEANING
Soundings used	Every ping, every 2nd (default), 3rd or 5th.

Returns	Strong only, medium and strong (default), or all above the sonar's low threshold. Soundings farther than 40 m are left out: at grazing angles a small sound-speed error moves the seabed by decimetres.

Result and Finalize

The table gives each correction with its uncertainty, or *not determined (0)*. The steps list how well the seabed agrees as recorded and after each correction kept. The values under **Values used** are what `FINALIZE` writes; type others to override. CloudPro uses them when it georeferences (`CONTINUE IN CLOUDPRO ›`). StripPro is for Lidar lines and does not take part in a marine job.

19 Reference

Conventions

AlignPro uses the mapper's own conventions, so its numbers go into the mapper unchanged.

- **Vehicle frame**: X forward, Y starboard, Z down, from the Certus reference point. Lever arms in metres, Forward / Starboard / Down.
- **Angles**: roll + starboard down, pitch + bow up, heading + clockwise, in degrees.
- **Lidar mounting**: M = Rz(heading) · Ry(pitch) · Rx(−roll) · Z0.
- **Near pitch 90°** (an upright mounting) roll and heading trade against each other: only their sum or difference is defined, so a tiny correction can change them a lot. The three angles together are still exact: enter all three. AlignPro warns and also gives the correction about the vehicle axes, which stays readable.
- **Errors**: mean point-to-plane distance over the planar patches, in centimetres, with the planes fitted again for each set of values.
- **SDEV**: the spread of the leave-one-line-out solutions (one sigma); with fewer than three used lines, the solver's formal standard deviation.

Settings

TAB	SETTING (KEY)	DEFAULT	NOTES
Data	Return used (`returns`)	strongest	strongest, last or both. One return per firing avoids near-duplicate points.
	Range min (m) (`range_min`)	1.5	
	Range max (m) (`range_max`)	60	Far points carry the most angle signal but also the most noise; 60 m suits the XT-32.
	Intensity min (`intensity_min`)	0	
	Points kept per line (`points_per_line`)	3,000,000	Random thinning budget; reads the lines again.
Passes and pairs	Split captures into passes at turns (`split_turns`)	on	
	Minimum speed (m/s) (`min_speed`)	1.0	Slower periods (statics) are dropped.
	Turn limit (deg) (`turn_deg`)	35	
	Minimum pass length (s) (`min_pass_s`)	4	
	Reciprocal pair: heading difference ≥ (deg) (`reciprocal_deg`)	150	
	Same direction: heading difference ≤ (deg) (`same_dir_deg`)	30	
	Minimum overlap (%) (`min_overlap_pct`)	20	
	Latency pair: speed difference ≥ (m/s) (`speed_margin`)	2.0	

Patches	Patch size (m) (`voxel_m`)	2.0	Voxel edge; 1.5–3 m for roadside scenes.
	Max thickness per line (m) (`thickness_m`)	0.04	A patch must be planar within this in every line that sees it.
	Min points per line (`min_pts_line`)	20	
	Max points per line per patch (`max_pts_patch_line`)	250	Set with the others by Point density.
	Flat patches per non-flat patch (`flat_ratio`)	1.0	Caps ground so walls drive heading.
	Max patches (`max_patches`)	8000	
	Solver memory limit (points) (`max_solve_points`)	12,000,000	Above this many patch points the solver keeps a random subset.
	Use only the area of interest (`use_aoi`)	off	Set by Draw area of interest and Clear area.
Solver	Huber threshold (sigma) (`huber_k`)	2.0	Down-weights residuals beyond this.
	Reject after convergence (sigma, 0 = off) (`reject_sigma`)	3.5	
	Max iterations (`max_iter`)	12	
	Also solve the Lidar lever arm (weak; for diagnosis) (`solve_lever`)	off	
Display	Grid cell size (m) (`grid_cell_m`)	0.5	Display only.

Data and pass settings read the lines again; the grid cell size only redraws the grid; the others find the pairs and patches again. Settings are saved per project.

Files

PATH IN `PROCESSING/<NAME>/02_ALIGNPRO/`	HOLDS
`alignpro.json`	AlignPro's settings, applied values, boxes, area of interest, last calibration result, report header and log (a `.bak` copy beside it)
`work/`	Line caches, the map grid, report screenshots; can be deleted (rebuilt on the next read)
`out/`	`alignment.json`, `alignment_report.txt`, `alignment_combinations.csv`, the PDF and text reports
`log/`	One log per run, e.g. `20261104_095507_auto_calibration.log`